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Analytical Problem Solving Assignment

Question: Solve the Water Jug problem: Using a 4-gallon jug and a 3-gallon jug, obtain exactly 2 gallons of water in the 4-gallon jug.

1. Problem Understanding (20%)

Given two unmarked jugs of capacities 4 gallons and 3 gallons. The allowed operations are: fill a jug completely, empty a jug, and pour water from one jug to the other until one is full or the other is empty. The objective is to obtain exactly 2 gallons in the 4-gallon jug.

2. Method Selection (20%)

State-space problem solving is used. Each state is represented as (4-gallon jug, 3-gallon jug). The solution is obtained by systematically applying the allowed operations until the goal state is reached.

3. Solution Process (25%)

4. Accuracy of Calculation (20%)

Final state = (2,0). Therefore, the 4-gallon jug contains exactly 2 gallons, satisfying the required condition.

5. Interpretation of Results (15%)

The problem is solved using only the permitted operations without any measuring device. The final goal state proves that analytical reasoning and systematic state transitions can solve the water jug problem efficiently.

Conclusion

The required quantity of 2 gallons is successfully obtained in the 4-gallon jug. This solution follows a logical sequence and satisfies all constraints given in the problem.

Question: A customer wants to travel from one location to another using the OLA Cab booking mobile application. The customer can select any cab type such as Mini, Micro, Sedan, Shared, Prime, etc. Identify the type of problem-solving agent used and write the pseudocode.

1. Problem Understanding (20%)

The customer enters the pickup and destination locations, views the available cab types, and selects a cab based on comfort, availability, and fare. The system must book the most suitable cab.

2. Method Selection (20%)

A Goal-Based Agent is the most appropriate because it selects the action that achieves the customer's goal of reaching the destination using the preferred cab.

3. Solution Process (25%)

Pseudocode:

```
START
Input Pickup_Location
Input Destination
Display Available_Cab_Types
User_Selects_Cab
IF Cab_Available THEN
    Calculate_Fare
    Confirm_Booking
    Assign_Driver
    Display_Driver_Details
    Start_Trip
ELSE
    Display "No Cab Available"
ENDIF
END
```

4. Accuracy of Calculation (20%)

The Goal-Based Agent correctly checks cab availability, calculates the fare, confirms the booking, and assigns a driver.

5. Interpretation of Results (15%)

The Goal-Based Agent successfully helps the customer reach the destination by selecting the most suitable cab. This improves efficiency and customer satisfaction.

Conclusion

The OLA cab booking application uses a Goal-Based Agent because it makes decisions based on achieving the customer's travel goal.

Question: A logistics company operates a delivery network where packages must be transported between warehouses at minimum cost. Determine the least-cost path from the starting warehouse S to the destination warehouse G using the Uniform Cost Search (UCS) algorithm.

1. Problem Understanding (20%)

The delivery network is represented as a weighted graph. Each edge has a travel cost. The objective is to find the least-cost path from S to G using Uniform Cost Search (UCS).

2. Method Selection (20%)

Uniform Cost Search (UCS) is selected because it always expands the node with the lowest cumulative path cost, guaranteeing the optimal path when all edge costs are positive.

3. Solution Process (25%)

Least-cost path found by UCS:

$S \rightarrow A \rightarrow C \rightarrow G$

Total Cost = $1 + 1 + 2 = 4$

4. Accuracy of Calculation (20%)

Possible paths:

- $S \rightarrow G = 12$
- $S \rightarrow A \rightarrow B \rightarrow D \rightarrow G = 1 + 3 + 3 + 3 = 10$
- $S \rightarrow A \rightarrow C \rightarrow D \rightarrow G = 1 + 1 + 1 + 3 = 6$
- $S \rightarrow A \rightarrow C \rightarrow G = 1 + 1 + 2 = 4$ (Minimum)

5. Interpretation of Results (15%)

UCS successfully identifies $S \rightarrow A \rightarrow C \rightarrow G$ as the optimal route with the minimum travel cost of 4. This minimizes transportation cost and improves delivery efficiency.

Conclusion

Uniform Cost Search guarantees the optimal solution for this weighted graph. The least-cost path is $S \rightarrow A \rightarrow C \rightarrow G$ with a total cost of 4.